



HR Analytics

Final Project



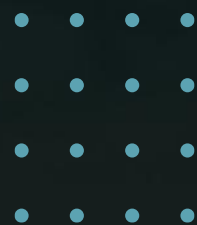
Name : Andrew Raafat



Introduction :

Unlocking Workforce Insights: A Data-Driven HR Strategy for 999 Employees.

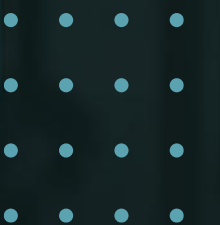
The goal is to extract insights, build predictive models, and support decision-making.

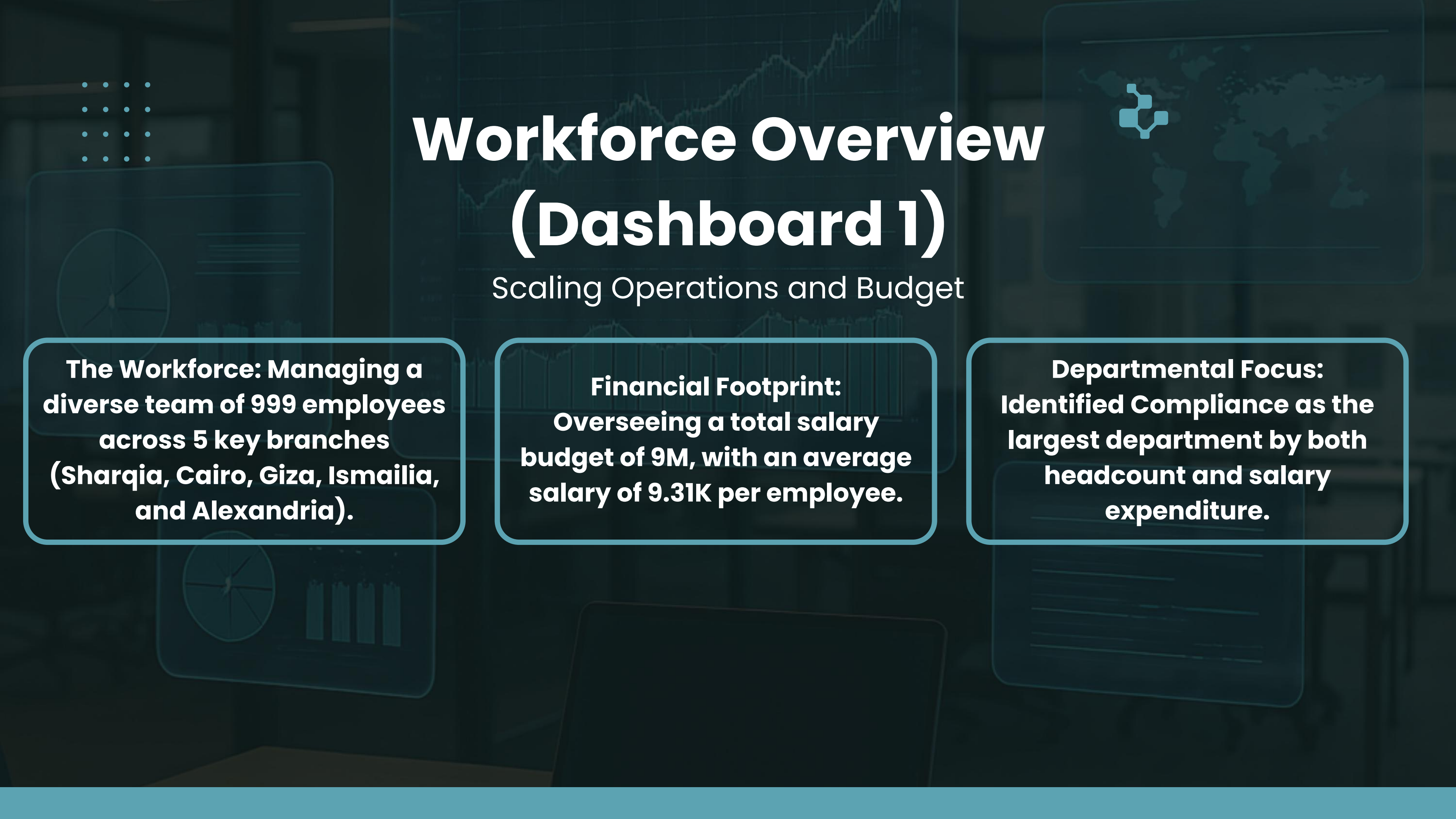




Data Cleaning & Preprocessing :

- Refining the Focus: Removed non-impactful identifiers like Names and IDs to ensure a bias-free analysis.
- Feature Engineering: Calculated "Age" and "Tenure" (Years of Experience) from date columns to identify the true drivers of salary.
- Standardization: Cleaned text entries and encoded categorical variables to prepare the dataset for high-accuracy modeling.





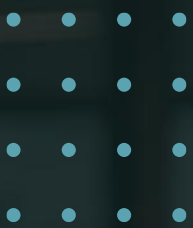
Workforce Overview (Dashboard 1)

Scaling Operations and Budget

The Workforce: Managing a diverse team of 999 employees across 5 key branches (Sharqia, Cairo, Giza, Ismailia, and Alexandria).

Financial Footprint: Overseeing a total salary budget of 9M, with an average salary of 9.31K per employee.

Departmental Focus: Identified Compliance as the largest department by both headcount and salary expenditure.



Demographics & Experience (Dashboard 2)

Stability Through Experience

Organizational Maturity: The company boasts a high level of loyalty, with an average tenure of 19.60 years.

Age Distribution: Our workforce peaks between the ages of 30 and 40, representing a stable and experienced professional core.

Hierarchy Insight: The structure consists of 671 Employees, 159 Supervisors, and 169 Managers.



Predictive Modeling :

90% Accuracy in Salary Forecasting

- The Solution: Implemented a Random Forest Regressor to decode the complex relationship between age, experience, and compensation.
- Model Performance: Achieved a high R^2 Score of 0.90, allowing the HR department to predict fair market salaries with 90% confidence.
- Strategic Value: This model serves as a benchmark for new hires and budget planning to ensure internal pay equity.

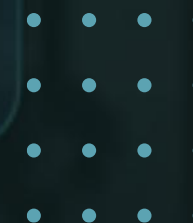




Conclusion & Recommendations :

Sustaining Growth

- Equity First: Use the 90% accurate prediction model to standardize salary offers and reduce turnover.
- Generational Balance: While our tenure is strong (19.6 years), we recommend targeted recruitment of younger talent to ensure a continuous transfer of knowledge.
- Efficiency: Focus on departments like Logistics, which show high efficiency with lower comparative salary costs.





Thank You



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